

Seowoo Jung

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EDUCATION

Ewha Womans University

Ph.D in Statistics, Advisor: Dr. Eun-Kyung Lee

Seoul, Republic of Korea

Mar 2023 – Feb 2027 (Expected)

Ewha Womans University

B.S. in Statistics, Minor in Computer Science

Seoul, Republic of Korea

Mar 2019 – Feb 2023

RESEARCH INTERESTS

Bayesian CART, Markov chain Monte Carlo, Causal inference, Machine Learning

RESEARCH EXPERIENCE

Research Assistant

Ewha Womans University (Advisor: Dr. Eun-Hee Ha)

Seoul, Republic of Korea

Nov 2022 – Feb 2024

- Machine learning based identification of factors associated with blood heavy metals in pregnant women.
- Mediation analysis of dietary patterns on depression via heavy metal exposure.
- Effectiveness and satisfaction of metaverse-based environmental medicine practical education.

Research Assistant

Ewha Womans University (Advisor: Dr. Eun-Kyung Lee)

Seoul, Republic of Korea

Mar 2023 – Present

- Developed an R package for Bayesian single-index models (BayesSIM).
- Developed a Bayesian projection pursuit tree regression model with interpretable nonlinear splitting rules.
- Analyze the effect of anesthetic drug using pharmacokinetic modeling.

PUBLICATIONS

Submitted

- Jung, S., Lee, E. K., “BayesSIM: An R Package for Bayesian Single-Index Model via nimble”.

Published

- Jung, S.*, Shah, S.*, Oh, J., Bang, Y., Lee, J. H., Kim, H. C., Jeong, K. S., Park, H., Lee, E. K., Hong, Y. C., Ha, E., Ko-CHENS Study group, “Machine Learning-Based Analysis on Factors Influencing Blood Heavy Metal Concentrations in the Korean Children’s Environmental Health Study (Ko-CHENS)” (2025), *Science of the Total Environment*, 978:179401.
- Jung, S., Kim, M., Lee, J., Min, J., Kim, H., Lee, E. K., Ha, E., “Analysis of the Association of Dietary Patterns With Depression in Pregnant Women Using Big Data – Mediating Effect of Blood Heavy Metal Concentration” (2025), *Journal of Health Informatics and Statistics*, 50(2):181–191.

- Shah, S., Oh, J., Bang, Y., **Jung, S.**, Kim, H. C., Jeong, K. S., Park, M. H., Lee, K. A., Ryoo, J. H., Kim, Y. J., Song, S., Park, H., Ha, E., Ko-CHENS study group, “Pregnant Women’s Lifestyles and Exposure to Endocrine-Disrupting Chemicals: A Machine Learning Approach” (2025), *Environmental Pollution*, 366:125309.

∗: Co-first author

RESEARCH PRESENTATIONS

Joint Statistical Meetings (JSM) 2026

Boston, USA
2026

- Contributed Session: A Bayesian Framework for Projection Pursuit Trees with Flexible Split Rules

Korean Statistical Society Conference

Gyeongju, South Korea
2025

- Poster: Investigating Robustness of G-Estimation in the Presence of Model Misspecification

useR!2024

Salzburg, Austria
2024

- Poster: Multilevel Regression and Poststratification with Projection Pursuit Tree

Korean Statistical Society Conference

Seoul, South Korea
2023

- Poster: Analysis of the Association of Dietary Patterns With Depression in Pregnant Women Using Big Data – Mediating Effect of Blood Heavy Metal Concentration

International Society for Environmental Epidemiology (ISEE)

Kaohsiung, Taiwan
2023

- Poster: Machine Learning-Based Analysis on Factors Influencing Blood Heavy Metal Concentrations in the Korean Children’s Environmental Health Study (Ko-CHENS)

AWARDS & HONORS

Best Paper Award

National Institute of Environmental Research

Seoul, Republic of Korea
2026

- Paper: Machine Learning-Based Analysis on Factors Influencing Blood Heavy Metal Concentrations in the Korean Children’s Environmental Health Study

AI Seoul Tech Graduate Scholarship

Seoul Future Foundation

Seoul, Republic of Korea
2025

- Graduate Fellowship for graduate student in Artificial Intelligence

Research Excellence Award

Ewha Womans University

Seoul, Republic of Korea
2025

- Topic: Extending Multilevel Regression and Poststratification with Bayesian Projection Pursuit Trees

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| Excellence Award, Big Data Analysis Contest (Bigcontest)
<i>Korea Association for ICT Promotion</i> | Seoul, Republic of Korea
2024 |
| <ul style="list-style-type: none"> • Topic: Analysis of Determinants of Housing Price Increases and Price Prediction in Seoul | |
| Summa Cum Laude, B.S. in Statistics
<i>Ewha Womans University</i> | Seoul, Republic of Korea
2023 |
| National Scholarship for Science and Engineering
<i>Korea Student Aid Foundation</i> | Seoul, Republic of Korea
2021–2022 |
| <ul style="list-style-type: none"> • Merit-based Full Tuition Waiver | |

TEACHING EXPERIENCE

- | | |
|--|---|
| Peer Instructor | 2022–Present
<i>Ewha Womans University</i> |
| <ul style="list-style-type: none"> • 2022–2023: Categorical Data Analysis • 2024, 2026: Statistical Programming 1: Assistance with R | |

TECHNICAL SKILLS

Advanced: Python, R
Intermediate: C, SAS, $\text{\LaTeX}$

REFERENCE

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Department of Statistics,
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